appearing periodically in the Journal of Physical and Chemical Reference Data.

B) If you are looking for a more complex specie or a radical, the available sources are the computerized databases of NIST Structures and Properties Database # 25 ver 2.0 1994, and the WebBook (Afeefy *et al*.), or periodic articles in the literature. There exists a database of molecular properties by Carl Melius that includes 3700 species. Many of these species are transition states. The Melius thermochemical tables are not available to the public, and only limited thermochemical information ($\Delta_f H_{300}$ and $\Delta_f G_{300}$) was published with his molecular structure data. The published thermo-chemical values calculated with the 1987 BAC/MP4 method are of low reliability.

C) For a solid or liquid species (not ideal gas) like $CaSO_4$ the place to look for is the JANAF compilation or the Barin (1995) compilation or the report of McBride, Gordon and Reno, NASA TM 4513 (1993).

D) Silicon containing species can be found in the CHEMKIN database (Kee *et al.*, (1992)). Sandia has initiated a database by M. Allendorf that includes presently Al, B, Si and Sn species calculated with the BAC/MP4 method. Other organometalic species of Ga and As can be found in an article by Tirtowidjojo and Pollard (1986).

E) Ion of simple bi- and triatomic species can be found in JANAF. The other compilation that mentions $\Delta_f H_{298}$ for ion species for some of the molecules and radicals included, is the above mentioned NIST S and P #25 computerized database (1994). The information therein was taken from Lias *et al*., Journal of Physical and Chemical Reference Data, Vol. 17, (1988), Supplement # 1.

Conversion Factors

The following conversion factors were used in the present compilation:

1 cal = 4.184 J

R = 8.314472 J mol^{-1} K^{-1}

R = 1.987207 cal mol^{-1} K^{-1}

1 eV = 23.06055 kcal mol^{-1}

1 eV = 8065.537 cm^{-1}

1 kcal/mol = 349.7547 cm^{-1}

1 cm^{-1} = 2.859146 cal mol^{-1}

1 cm^{-1} = 29.9792458 GHz

1 Hartree = 627.5101 kcal mol^{-1}

1 Hartree = 2625.502 kJ mol^{-1}

1 Bohr = 0.5291772 Å

1×10^{-39} g cm^2 = 21.50545 amu Bohr^2

1×10^{-39} g cm^2 = 6.022137 amu Å^2

B/cm^{-1} = 60.19969/[I_B/(amu Bohr^2)]

B/cm^{-1} = 16.85763/[I_B/(amu Å^2)]

B/cm^{-1} = 2.799277/[I_B/(10^{-39} g cm^2)]

Electronic Files

Updated version of the database in ASCII form (BURCAT.THR), is available for free downloading from:
ftp://ftp.technion.ac.il/pub/supported/aetdd/thermodynamics.
The site is mirrored daily by:
http://garfield.chem.elte.hu/Burcat/burcat.html
Transfer the file to your computer using download, or browse through it with your web browser.

Two table generator programs CAP and CAPOLD written by B. McBride are included, to enable generation of thermodynamic properties from the given nine term and seven term polynomials, respectively. The tables generated provide values of C_p, S, H_T-H_{298}, -(G_T-H_{298})/T and H_T as a function of T. Values of $\Delta_f H_T$ and log K_c can be added at any temperature interval within the polynomial's given limit.

A special file called THERM. DAT contains the file burcat.thr stripped of all comments, to be used with

```
53590-28-6
C3H RAD  CC-CH  T0=0  STATWT=2.  SIGMA=1.  IA=0.0353 IB=7.5023  IC=7.5376
Nu=3238,1825,1167,467,[72.5,275.5]  T0=19187  STATWT=2  IA=0.0353 IB=7.5023
 IC=7.5376  Nu=2800,1836,1091,881,460(2)  T0=20538. STATWT=2 IA=0.0353 IB=7.5023
 IC=7.5376   Nu=2800,1836,1091,784,493(2)   HF298=171.94+/-1.9 kcal  HF0=170.67
kcal  REF=Burcat G3B3 calc; Vibrations from Jacox (Webbook 2005) and G3B3 calc[]
{HF0=127.1 kcal  REF=Duff & Bauer Los Alamos Rep 2556 1961; SPANGENBERG &
BORGER  Z.Phys. Chem (Leipzig) 255,(1974),1; HF298=163.5 kcal  REF=Estimated
from C2H C4H and C6H by Kiefer et al Comb. Sci Technol 82,(1992),101  *** Note
Duff & Bauer expect SIGMA=4 and Spangenberg gives SIGMA=3 for HCC-C linear
configuration}   MAX LST SQ ERROR Cp @ 1300K 0.3%.
C3H  Radical HCCC A 7/05C  3.H  1.   0.   0.G   200.000  6000.000  B  37.04004 1
 6.14184491E+00 3.39661013E-03-1.21915444E-06 1.97782838E-10-1.18312807E-14    2
 8.44225753E+04-6.44480148E+00 3.34917187E+00 1.65822626E-02-2.77115653E-05    3
 2.51382364E-08-8.85285352E-12 8.49863168E+04 6.80362439E+00 8.65225703E+04    4

431-89-0
C3HF7 2-HEPTAFLUORO-PROPANE CF3-CHF-CF3 (FC-227ea) SIGMA=2 STATWT=1  IA=38.9667
IB=78.08525  IC=88.49144  NU=2966,1428,1392,1324,1296,1280,1242,1224,1150,1131,
896,853,725,669,593,535,519,502,442,335,314,283,232,214,152,89.7.26.9
HF298=-374.00  KJ REF=Melius Database 1987 AB1V  {HF298=-374.47 kcal  REF=Zhang
JOC 63,(1998),3590-94}  Max Lst Sq Error Cp @  1300 K 0.44%.
C3F7H  FC227EA    T12/99C  3.H  1.F  7.   0.G   200.000  6000.000  C 170.02886 1
 2.03195617E+01 1.04618873E-02-3.99351610E-06 6.70976809E-10-4.12886922E-14    2
-1.96070480E+05-7.39087817E+01 3.19381844E+00 5.64358210E-02-4.24435538E-05    3
 6.01422805E-09 4.21730731E-12-1.91302556E+05 1.47970140E+01-1.88203033E+05    4

1070-71-9
C3HN CYANO-ACETYLENE HCC-CN  SIGMA=1  STATWT=1  IB=18.4925   Nu=3489,2385,2186,
911,651(2),573.5(2),260(2)  HF298=368.414 kJ  HF0=367.225 kJ   REF=Burcat G3B3
calc  {HF298=84.6 kcal REF=ESTIMATED BY MACKIE & COLKET 22nd COMBUST SYMP 1990;
HF298=84.0 kcal REF=Knight Freeman McEwan Int.J Mass. Spect.Ion Phys. 67,(1985),
317; HF298=90.7 kcal  NIST 94}    Max Lst Sq Error Cp @ 1300 K 0.34%
C3HN Cyano-Acety  A 2/05C  3.H  1.N  1.   0.G   200.000  6000.000  B  51.04678 1
 7.44515032E+00 5.27107604E-03-1.86735278E-06 2.98683734E-10-1.77665376E-14    2
 4.16450237E+04-1.46187448E+01 5.87779106E-01 3.84323486E-02-6.61566501E-05    3
 5.72555769E-08-1.89892637E-11 4.29066005E+04 1.74909167E+01 4.43097371E+04    4

16165-40-5
C3H2 CYCLOPROPENYLIDENE BI-RADICAL SINGLET SIGMA=2 STATWT=1 Ia=2.35340 Ib=2.4065
Ic=4.8941  Nu=788,887,[898,979],1063,1277,[1588,3080,3114] REF=Webbook NIST2000
+[]Vereecken, et al JCP 108,(1998),1068   HF298=114 kcal  REF= Kiefer et.al.
J. Phys. Chem 101, (1997), 4057   {HF298=121.63+/-6.3 kcal REF=Melius 1988 P60V}
{HF298=136 kcal   REF=PM3 RHF calculation}   Max Lst Sq  Error Cp @ 200 K 0.82%
C3H2(1) Cyclo     T12/00C  3.H  2.   0.   0.G   200.000  6000.000  B  38.04888 1
 5.69445684E+00 6.53821901E-03-2.35907266E-06 3.82037384E-10-2.29227460E-14    2
 5.49264274E+04-6.96163733E+00 3.18167129E+00-3.37611741E-04 3.95343765E-05    3
-5.49792422E-08 2.28335240E-11 5.61816758E+04 9.06482468E+00 5.73666999E+04    4
```

```
C3H  Radical CC-CH  HF298=171.94 kcal   Burcat G3B3 calc
 3 A 7/05 C   3.00H   1.00    0.00    0.00    0.00 0   37.0400400     719392.776
     50.000   200.000 7 -2.0 -1.0  0.0  1.0  2.0  3.0  4.0  0.0        12695.889
-3.759807260D+03 2.371933767D+02-1.291060829D+00 4.379555470D-02-2.775884627D-05
-3.962948530D-07 9.081247010D-10 0.000000000D+00 8.420418960D+04 2.784182496D+01
    200.000  1000.000 7 -2.0 -1.0  0.0  1.0  2.0  3.0  4.0  0.0        12695.889
-7.996695870D+03-1.576691670D+02 6.013789640D+00 3.627872400D-03-2.349637630D-07
-1.285382488D-09 5.558659370D-13 0.000000000D+00 8.544417940D+04-6.096137210D+00
   1000.000  6000.000 7 -2.0 -1.0  0.0  1.0  2.0  3.0  4.0  0.0        12695.889
 1.060461380D+06-4.270459160D+03 1.229408959D+01-6.623009490D-04 9.393153300D-08
-4.148656840D-12-5.784616800D-17 0.000000000D+00 1.104688155D+05-4.921984460D+01

CF3CHFCF3   MELIUS Molec 48 in MP2.97y  HF298(CBS-4)=374.47 kcal
 3 T12/99 C   3.00H   1.00F   7.00    0.00    0.00 0  170.0288624   -1564816.000
     50.000   200.000 7 -2.0 -1.0  0.0  1.0  2.0  3.0  4.0  0.0        25901.254
-1.128401047D+03 1.070136559D+02 6.141693160D-01 7.910789350D-02-1.243096100D-04
 1.102068219D-07-3.653663810D-13 0.000000000D+00-1.916362533D+05 2.581048163D+01
    200.000  1000.000 7 -2.0 -1.0  0.0  1.0  2.0  3.0  4.0  0.0        25901.254
-1.661680469D+05 2.765832248D+03-1.473016441D+01 1.140529546D-01-1.389340054D-04
 8.596830610D-08-2.155059484D-11 0.000000000D+00-2.041287011D+05 1.117173147D+02
   1000.000  6000.000 7 -2.0 -1.0  0.0  1.0  2.0  3.0  4.0  0.0        25901.254
-9.268076300D+04-6.443414210D+03 3.517541420D+01-1.511748262D-03 3.108954389D-07
-3.381205300D-11 1.505926580D-15 0.000000000D+00-1.617880777D+05-1.729691122D+02

C3HN Cyano-Acetylene HCC-CN  Burcat G3B3 calc. HF298=88.053 kcal
 3 A 2/05 C   3.00H   1.00N   1.00    0.00    0.00 0   51.0467800     368413.752
     50.000   200.000 7 -2.0 -1.0  0.0  1.0  2.0  3.0  4.0  0.0        12918.213
 2.937389551D+02 3.506911300D+01 1.262065637D+00 3.591227570D-02-1.583680848D-04
 7.254716590D-07-1.277955096D-09 0.000000000D+00 4.269771270D+04 1.517037217D+01
    200.000  1000.000 7 -2.0 -1.0  0.0  1.0  2.0  3.0  4.0  0.0        12918.213
 1.790375913D+05-2.774379850D+03 1.741345965D+01-1.257670664D-02 1.519433093D-05
-7.549862690D-09 1.248968939D-12 0.000000000D+00 5.596479150D+04-7.454819530D+01
   1000.000  6000.000 7 -2.0 -1.0  0.0  1.0  2.0  3.0  4.0  0.0        12918.213
 1.580169162D+06-6.500524190D+03 1.690821764D+01-1.010038617D-03 1.721322862D-07
-1.570147037D-11 5.931071320D-16 0.000000000D+00 8.120065870D+04-8.033854750D+01

C3H2F3  CF3-CH=CH*  Radical  Burcat G3B3 calc  HF298=-90.080 kcal
 3 A10/04 C   3.00H   2.00F   3.00    0.00    0.00 0   95.0431896    -376894.720
     50.000   200.000 7 -2.0 -1.0  0.0  1.0  2.0  3.0  4.0  0.0        17441.709
 8.631818300D+02-7.716526700D+01 6.332642530D+00-2.817025687D-02 3.786899150D-04
-1.174123246D-06 1.391322805D-09 0.000000000D+00-4.719591440D+04 1.750004464D+00
    200.000  1000.000 7 -2.0 -1.0  0.0  1.0  2.0  3.0  4.0  0.0        17441.709
 4.676745000D+04-2.247301983D+02-9.803873070D-01 6.059522790D-02-8.251558710D-05
 5.709019040D-08-1.576255475D-11 0.000000000D+00-4.566980150D+04 2.908334754D+01
   1000.000  6000.000 7 -2.0 -1.0  0.0  1.0  2.0  3.0  4.0  0.0        17441.709
 9.372963500D+05-6.535012900D+03 2.439446351D+01-5.848392840D-04 3.577943500D-08
 1.445940747D-12-2.066763842D-16 0.000000000D+00-1.237160688D+04-1.170715520D+02
```